

Problem Set 9

A. Constellation

tianzheyu submissions

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#	When	Who	Problem	Lang	Verdict	Time	Memory
385837035	Aug/06/2026 15:37UTC+10	tianzheyu	C - Constellation	C++23 (GCC 14-64, msys2)	Accepted	390 ms	100 KB

Process

Sorted the stars by their x and y coordinates, selected the first two points, and then iterated through the rest to find the first point that is not collinear with them.

Challenges and Overcoming

When coding the condition to find the third point, I ran into a logic bug where I mistakenly used `==` instead of `!=` for the cross-product check, causing the code to look for collinear points instead of a valid triangle.

Additionally, when calculating the cross product, I ran into a bug where I used standard 32-bit `int` data types for the coordinates. I found it when my submission crashed on Test #93 with a `runtime error: signed integer overflow`. Since the coordinates can be up to 10^9 , multiplying their differences exceeded the maximum limit of a 32-bit integer. Next time to prevent this bug, I will always double-check the variable constraints in the problem description before coding and default to using `long long` for any geometric calculations involving multiplication.